

SOFTWARE TESTING AND QUALITY ASSURANCE

Mini E-Commerce Website – Test Case Document

Comprehensive Functional, Integration, System, Boundary, Security and Negative Test Suite

Project	Mini E-Commerce Website
Subject	Software Testing and Quality Assurance (STQA)
Test Cases	160
Application Areas	Authentication, Registration, Products, Cart, Checkout, Orders, Payments, Inventory, APIs, Authorization, Security
Execution Fields	Actual Result and Status intentionally left blank for lab execution

Testing Objectives

- Verify that the major e-commerce modules work according to their expected behavior.
- Identify invalid-input, boundary, negative and authorization failures.
- Verify interactions between modules such as products, cart, checkout, payments, orders and inventory.
- Provide a comprehensive test suite suitable for STQA laboratory execution and viva.

1. Authentication & Login

TC ID	Test Scenario	Test Data	Expected Result	Actual Result	Status
TC001	Login with valid customer credentials	Valid customer email + password	Customer dashboard displayed	—	—
TC002	Login with valid admin credentials	Valid admin email + password	Admin dashboard displayed	—	—
TC003	Invalid email	Wrong email + valid password	Login rejected	—	—
TC004	Invalid password	Valid email + wrong password	Login rejected	—	—
TC005	Both credentials invalid	Wrong email + password	Login rejected	—	—
TC006	Empty email	Email = blank	Validation error	—	—
TC007	Empty password	Password = blank	Validation error	—	—
TC008	Both fields empty	Blank + blank	Validation errors	—	—
TC009	Invalid email format	abc@	Validation error	—	—
TC010	Password below minimum length	Short password	Validation error	—	—
TC011	Logout	Logged-in user	User logged out	—	—
TC012	Access protected page after logout	Logged-out user	Redirected or access denied	—	—
TC013	Expired/invalid JWT	Invalid token	Request rejected	—	—
TC014	Login with disabled account	Disabled customer account	Login rejected	—	—
TC015	Repeated failed login attempts	Multiple wrong passwords	Account/rate-limit protection triggered	—	—

2. Customer Registration

TC ID	Test Scenario	Test Data	Expected Result	Actual Result	Status
TC016	Register valid customer	All valid fields	Customer created	—	—
TC017	Duplicate email registration	Existing email	Registration rejected	—	—
TC018	Invalid email	abc.com	Validation error	—	—
TC019	Empty name	Name blank	Validation error	—	—
TC020	Empty email	Email blank	Validation error	—	—
TC021	Empty password	Password blank	Validation error	—	—
TC022	Invalid phone number	Alphabetic phone	Validation error	—	—
TC023	Short phone number	123	Validation error	—	—
TC024	Valid phone number	10-digit number	Accepted	—	—
TC025	Special characters in name	@#\$\$	Validation handled	—	—
TC026	Very long name	Above allowed length	Validation/error handled	—	—

TC ID	Test Scenario	Test Data	Expected Result	Actual Result	Status
TC027	Duplicate phone number	Existing phone	Registration rejected	—	—

3. Product Browsing & Search

TC ID	Test Scenario	Test Data	Expected Result	Actual Result	Status
TC028	View product listing	Available products	Products displayed	—	—
TC029	View product details	Existing product	Correct details displayed	—	—
TC030	Search existing product	Existing product name	Matching products displayed	—	—
TC031	Search nonexistent product	Unknown product	No matching result	—	—
TC032	Search by category	Existing category	Category products displayed	—	—
TC033	Filter by price range	Valid min/max price	Matching products displayed	—	—
TC034	Sort by price ascending	Product list	Lowest price first	—	—
TC035	Sort by price descending	Product list	Highest price first	—	—
TC036	Filter with no matching products	Unmatched criteria	No products displayed	—	—
TC037	View out-of-stock product	Out-of-stock product	Stock status shown	—	—
TC038	Open invalid product ID	Nonexistent ID	Appropriate error displayed	—	—
TC039	Special characters in search	<script>	Input handled safely	—	—

4. Product Management / Admin

TC ID	Test Scenario	Test Data	Expected Result	Actual Result	Status
TC040	Add product with valid data	Valid product data	Product created	—	—
TC041	Add product with duplicate SKU	Existing SKU	Duplicate rejected	—	—
TC042	Add product without name	Name blank	Validation error	—	—
TC043	Add product without price	Price blank	Validation error	—	—
TC044	Add product with negative price	Price < 0	Rejected	—	—
TC045	Add product with zero stock	Stock = 0	Accepted/handled according to rule	—	—
TC046	Add product with negative stock	Stock < 0	Rejected	—	—
TC047	Edit product	Valid updated data	Details updated	—	—
TC048	Deactivate product	Existing active product	Product becomes inactive	—	—
TC049	Delete product	Existing product	Product removed/disabled	—	—
TC050	Customer attempts product management	Customer account	Access denied	—	—

5. Shopping Cart

TC ID	Test Scenario	Test Data	Expected Result	Actual Result	Status
TC051	Add valid product to cart	Available product	Product added	—	—
TC052	Add same product twice	Same product twice	Quantity updated correctly	—	—
TC053	Add unavailable product	Out-of-stock product	Addition rejected	—	—
TC054	Update cart quantity	Valid quantity	Cart updated	—	—
TC055	Set cart quantity to zero	Quantity = 0	Item removed/handled	—	—
TC056	Set negative quantity	Quantity = -1	Validation error	—	—
TC057	Exceed available stock	Quantity > stock	Operation rejected	—	—
TC058	Remove product from cart	Existing cart item	Item removed	—	—
TC059	View empty cart	No cart items	Empty-cart message displayed	—	—
TC060	Calculate cart subtotal	Multiple valid items	Correct subtotal	—	—
TC061	Cart persists after refresh	Existing cart	Cart retained	—	—
TC062	Cart isolated between users	Two customer accounts	Each sees own cart	—	—

6. Checkout & Order Workflow

TC ID	Test Scenario	Test Data	Expected Result	Actual Result	Status
TC063	Checkout with valid cart	Valid cart + address	Checkout page displayed	—	—

TC ID	Test Scenario	Test Data	Expected Result	Actual Result	Status
TC064	Checkout empty cart	Empty cart	Checkout blocked	—	—
TC065	Missing shipping address	Address blank	Validation error	—	—
TC066	Invalid postal code	Invalid postal code	Validation error	—	—
TC067	Place order with valid data	Valid cart + address + payment	Order created	—	—
TC068	Place order with invalid payment	Invalid payment data	Order rejected	—	—
TC069	Place order with insufficient stock	Stock lower than cart quantity	Order rejected	—	—
TC070	Order total calculation	Multiple items + tax/shipping	Correct total	—	—
TC071	Apply valid coupon	Valid coupon	Discount applied	—	—
TC072	Apply invalid coupon	Invalid coupon	Coupon rejected	—	—
TC073	Apply expired coupon	Expired coupon	Coupon rejected	—	—
TC074	Apply coupon below minimum order	Order below threshold	Coupon not applied	—	—
TC075	Cancel pending order	Existing pending order	Order cancelled	—	—
TC076	Cancel delivered order	Delivered order	Cancellation rejected	—	—
TC077	View order details	Existing order	Correct order displayed	—	—
TC078	View order history	Logged-in customer	Correct orders displayed	—	—

7. Payment Testing

TC ID	Test Scenario	Test Data	Expected Result	Actual Result	Status
TC079	Pay exact order amount	Payment = order total	Payment successful	—	—
TC080	Pay less than order amount	Payment < total	Payment rejected/partial handled	—	—
TC081	Pay more than order amount	Payment > total	Validation/error handled	—	—
TC082	Invalid payment method	Unsupported method	Payment rejected	—	—
TC083	Payment gateway timeout	Gateway timeout	Order/payment handled safely	—	—
TC084	Payment failure	Gateway returns failure	Order not incorrectly marked paid	—	—
TC085	Successful payment updates order	Successful payment	Order status updated	—	—
TC086	Duplicate payment callback	Same callback twice	Duplicate charge/order avoided	—	—
TC087	Refund paid order	Paid order	Refund/status handled	—	—
TC088	Refund unpaid order	Unpaid order	Refund rejected	—	—

8. Order Management / Admin

TC ID	Test Scenario	Test Data	Expected Result	Actual Result	Status
TC089	Admin views all orders	Admin account	Orders displayed	—	—
TC090	Customer views own orders	Customer account	Own orders displayed	—	—
TC091	Customer accesses another user's order	Different order ID	Access denied	—	—
TC092	Admin updates order status	Valid order + status	Status updated	—	—
TC093	Invalid order status transition	Unsupported transition	Rejected/handled	—	—
TC094	Mark order shipped	Paid/confirmed order	Status becomes shipped	—	—
TC095	Mark order delivered	Shipped order	Status becomes delivered	—	—
TC096	Cancel already cancelled order	Cancelled order	Appropriate error	—	—
TC097	Invalid order ID	Nonexistent ID	Appropriate error	—	—
TC098	Modify URL order ID	Another customer's order ID	Access denied	—	—

9. Inventory / Stock

TC ID	Test Scenario	Test Data	Expected Result	Actual Result	Status
TC099	Increase product stock	Positive quantity	Stock increases correctly	—	—
TC100	Decrease product stock	Valid quantity within stock	Stock decreases correctly	—	—
TC101	Deduct more than available stock	Quantity > stock	Operation rejected	—	—
TC102	Stock reaches zero	Final available unit sold	Stock becomes 0	—	—
TC103	Low-stock condition	Stock below threshold	Low-stock alert generated	—	—
TC104	Restock product	Valid restock quantity	Stock updated	—	—

TC ID	Test Scenario	Test Data	Expected Result	Actual Result	Status
TC105	Negative restock quantity	Quantity < 0	Rejected	—	—
TC106	Concurrent purchase of last item	Two buyers, one unit	Only valid stock allocation succeeds	—	—
TC107	Inactive product purchase	Inactive product	Purchase rejected	—	—
TC108	Inventory after cancellation	Cancelled order	Stock restored according to rule	—	—

10. API & Integration Testing

TC ID	Test Scenario	Test Data	Expected Result	Actual Result	Status
TC109	GET products API	Valid GET request	200 response + product data	—	—
TC110	GET product by ID API	Valid product ID	200 response + correct product	—	—
TC111	POST login API	Valid credentials	200/appropriate success response + token	—	—
TC112	POST registration API	Valid customer data	Customer created	—	—
TC113	POST cart API	Valid product + quantity	Cart updated	—	—
TC114	POST order API	Valid checkout data	Order created	—	—
TC115	Payment API integration	Valid payment request	Payment result returned correctly	—	—
TC116	Database order persistence	Successful order API	Order stored correctly	—	—
TC117	Frontend-to-product API integration	Open product page	Correct API data rendered	—	—
TC118	Frontend-to-order API integration	Place valid order	UI reflects created order	—	—

11. Authentication / Authorization / RBAC

TC ID	Test Scenario	Test Data	Expected Result	Actual Result	Status
TC119	Customer accesses admin API	Customer JWT + admin endpoint	Access denied	—	—
TC120	Customer modifies product	Customer account	Denied	—	—
TC121	Customer modifies another user's cart	Different user ID	Access denied	—	—
TC122	Admin accesses customer management	Admin account	Access allowed	—	—
TC123	Unauthenticated API request	No JWT/token	401 or access denied	—	—
TC124	Invalid JWT	Malformed token	Request rejected	—	—
TC125	Expired JWT	Expired token	Request rejected	—	—
TC126	Access another user's private data	Different user ID	Denied	—	—
TC127	Modify URL ID to another user's order	Unauthorized order ID	Access denied	—	—
TC128	Reuse token after logout	Logged-out token	Token rejected/invalidated	—	—

12. Input Validation & Error Handling

TC ID	Test Scenario	Test Data	Expected Result	Actual Result	Status
TC129	Product name at minimum length	Minimum allowed length	Accepted	—	—
TC130	Product name one below minimum	Minimum - 1	Rejected	—	—
TC131	Product name at maximum length	Maximum allowed length	Accepted	—	—
TC132	Product name above maximum	Maximum + 1	Rejected	—	—
TC133	Quantity = 0	0	Rejected/handled	—	—
TC134	Quantity = 1	1	Accepted	—	—
TC135	Quantity = maximum	Maximum allowed	Accepted	—	—
TC136	Quantity = maximum + 1	Maximum + 1	Rejected	—	—
TC137	Price = 0	0	Rejected/handled	—	—
TC138	Price = negative	-1	Rejected	—	—
TC139	Price with excessive decimals	Too many decimal places	Validation error/handled	—	—
TC140	Malformed JSON request	Invalid JSON body	400/error response	—	—
TC141	Missing required API field	Required field omitted	Validation error	—	—
TC142	Unsupported HTTP method	Unsupported method	405/error response	—	—

13. Boundary Value Analysis

TC ID	Test Scenario	Test Data	Expected Result	Actual Result	Status
TC143	Minimum valid password length	Minimum allowed length	Accepted	—	—
TC144	One character below password minimum	Minimum - 1	Rejected	—	—
TC145	Maximum allowed name length	Maximum allowed length	Accepted	—	—
TC146	One character above maximum name length	Maximum + 1	Rejected	—	—
TC147	Cart quantity = 0	0	Rejected/handled	—	—
TC148	Cart quantity = 1	1	Accepted	—	—
TC149	Cart quantity = maximum	Maximum allowed	Accepted	—	—
TC150	Cart quantity = maximum + 1	Maximum + 1	Rejected	—	—
TC151	Coupon minimum order exactly met	Exact threshold	Coupon accepted	—	—
TC152	Coupon order one unit below threshold	Threshold - 1	Coupon rejected	—	—
TC153	Stock = 0	0	Out-of-stock condition	—	—
TC154	Stock = 1	1	Low-stock logic checked	—	—

14. Security & Regression Testing

TC ID	Test Scenario	Test Data	Expected Result	Actual Result	Status
TC155	SQL injection in search	' OR 1=1 --	Input rejected/safely handled	—	—
TC156	XSS payload in product name	<script>alert(1)</script>	Payload escaped/rejected	—	—
TC157	XSS payload in search		Payload safely handled	—	—
TC158	Brute-force login protection	Many failed attempts	Rate limit/lockout triggered	—	—
TC159	Sensitive password exposure	Login/API response	Password not exposed	—	—
TC160	Customer accesses admin endpoint	Customer token	Access denied	—	—

Testing Techniques Covered

- Functional Testing
- Negative Testing
- Boundary Value Analysis
- Equivalence Partitioning
- Decision-oriented testing of workflows
- Integration Testing
- System Testing
- Regression Testing
- API Testing
- Authentication Testing
- Authorization / Role-Based Access Control Testing
- Input Validation Testing
- Database / Data Integrity Testing
- Error Handling Testing
- Security-oriented Testing

Lab Execution Notes

- Execute the test cases against the locally configured e-commerce application.
- Do not pre-fill Actual Result or Status with assumed values; record the observed result during execution.
- Use Pass when actual behavior matches the expected result; otherwise use Fail and record the observed behavior.
- For failed cases, capture screenshots or API responses as evidence where appropriate.
- For API testing, record HTTP status codes and important response messages in the Actual Result field.